

RoboLetics 3.0: Benchmarks for Agile Physical Human-Robot Interaction

Workshop

<https://roboletics.github.io/corl2026/>

Organizers

Name	Affiliation	Email
Matthew Gombolay	Georgia Tech	matthew.gombolay@cc.gatech.edu
Andrew Morgan	Robotics and AI Institute	andy@rai-inst.com
Varshith Sreeramdas	Georgia Tech	vsreeramdas@gatech.edu
Kin Man Lee	Georgia Tech	klee863@gatech.edu
Zixuan Wu	Georgia Tech	zwu380@gatech.edu
Qingyu Xiao	Georgia Tech	qxiao33@gatech.edu
Kiersten Lange	Georgia Tech	klange31@gatech.edu
Dylan Barnes	Georgia Tech	dbarnes47@gatech.edu
Zulfiqar Zaidi	Georgia Tech	zzaidi8@gatech.edu

Description

Sports have served as the ultimate stress test for human agility and collective intelligence. As robots transition into shared everyday spaces, operating in proximity to humans necessitates approachable embodiments with high levels of autonomy and coordination. The pursuit of mixed human-robot sports provides a natural stepping-stone to developing robot partners that are simultaneously agile, collaborative, and trustworthy. Beyond technical advancement, sports provide an accessible and compelling medium to engage society with robotics research.

RoboLetics 3.0, champions mixed human-robot sports as the next grand challenge for robotics. We will explore the development of benchmarks based on collaborative physical sports to emphasize human robot interaction (HRI). By imposing physical constraints alongside real-time social dynamics, these benchmarks will provide structured environments to force the integration of embodiment design, explosive motor control and partner models, to develop safe, fluid, performant collaboration.

Our workshop brings together robot learning and HRI communities to collectively address critical challenges in agile physical HRI. Through a diverse program including invited talks from academic and industry experts, moderated audience discussions, spotlight presentations from early-career researchers, curated robot demonstration videos, a simulation benchmark prototype based on doubles tennis, structured breakout sessions, and a panel discussion, the workshop aims to synthesize perspectives from researchers at all career stages and diverse backgrounds. In partnership with the Robotics and AI Institute, our workshop will showcase the latest innovations, foster interdisciplinary dialogue, and develop a roadmap to guide the future of agile, collaborative robotics.

Core Challenges / Research Questions

Our workshop proposes mixed human-robot sports as a principled design space for benchmarking and catalyzing agile physical human-robot interaction research. Our goal is to identify benchmark tasks, evaluation methodologies, safety constraints, fairness criteria, rules for difficulty level progression that stress-test aspects of agile physical HRI, embodiment design, models, control algorithms and push their translation into real-world applications.

- Mapping sports properties to benchmark design: How to systematically translate sports’ characteristics into benchmark specifications? For instance, tasks with central objects in play (e.g., soccer, basketball, tennis) test agility and collaborative strategy, physical contact collaboration (e.g., cheerleading, acrobatics) balances coordination with safety, etc.
- Stress-testing algorithms and models: How to design benchmarks that effectively challenge models and algorithms for control involving dynamic contact, real-time safety verification, verbal/non-verbal communication, intent estimation, initiative-taking and role switching, human partner modeling, theory-of-mind reasoning?
- Progressive specification: How to formulate sports’ rules that enable systematic progression across difficulty levels so that it provides meaningful incremental challenges? How to design benchmarks that test collaboration and strategic consensus at various timescales — from real-time coordination to multi-step strategic planning?
- Multi-faceted safety constraints: How to formally specify and evaluate diverse safety dimensions such as physical safety during contact or psychological/perceived safety for human partners?
- Fairness and robot morphologies: How to define fair evaluation when robots have different morphologies, identify which have systematic advantages/disadvantages in specific sports, and design handicapping mechanisms that provide insightful comparisons across embodiments?
- Research prioritization: Which research questions should be prioritized in the short and long term to advance the field of agile physical HRI through sport-based benchmarking?

Invited Speakers

Nico Rojas

COMMITTED

Robotics and AI Institute

nrojas@rai-inst.com

Dr. Nico Rojas brings expertise in the mechanical design of high-performance robotic systems as a project lead for dynamic manipulation at RAI. His perspective will help the workshop determine how morphology, actuation, inertia, power, and mechanical robustness should be represented in sport-based benchmarks, including how embodiment enables athletic performance and how evaluation can remain meaningful and fair across heterogeneous robot platforms.

Guanya Shi

IN-PERSON

Carnegie Mellon University

guanyas@andrew.cmu.edu

Prof. Guanya Shi develops learning and control methods for reliable, adaptive, and agile general-purpose robots, with recent work on humanoid whole-body control and loco-manipulation. His expertise is directly relevant to designing benchmarks that couple locomotion, object interaction, contact, and rapid adaptation, and to defining progressive task levels that distinguish motion imitation from robust, physically grounded, whole-body athletic behavior.

Nadia Figueroa

REQUESTED

University of Pennsylvania

nadiafig@seas.upenn.edu

Prof. Nadia Figueroa studies fluid human-robot collaborative autonomy through tightly integrated learning, control, perception, and estimation methods for safe and adaptive physical interaction. Her perspective will help shape benchmarks that evaluate how robots respond to human motion, intent, and changing roles during dynamic collaboration, while incorporating interaction quality, safety, and robustness alongside conventional task-performance metrics.

Aaron Ames

REQUESTED

California Institute of Technology

ames@cds.caltech.edu

Prof. Aaron Ames develops control-theoretic methods for dynamic locomotion and safety-critical autonomy, with extensive validation on bipedal robotic systems. His expertise will help shape sport-based benchmarks that evaluate safety constraint enforcement, controller intervention, robustness to disturbances, and the balance between athletic performance and safe operation.

Proposed Format and Interactivity

The structure of the workshop reflects our objective to produce a community roadmap to collaboratively guide the future of robotics research. We provide several avenues to gather viewpoints from researchers with diverse backgrounds across all stages in their careers.

Time	Talk	Comments
8:30–8:40am	Introduction	Matthew Gombolay
8:40–9:00am	Talk 1	
9:00–9:20am	Talk 2	
9:20–9:30am	Moderated Audience Discussion	
9:30–9:50pm	Talk 3	
9:50–10:30pm	Student Paper Spotlights	
10:30–11:00am	Poster Session, Curated Demo Videos	Coffee Break
11:00–11:20am	Talk 4	
11:20–11:30am	Moderated Audience Discussion	
11:30–12:00pm	Breakout Session	
12:00–12:30pm	Panel Discussion	

Table 1: Event Schedule

Structured Breakout Sessions: Participants will break out into groups of 5–7. The breakout session will be organized around different axes of agile physical HRI that arise across multiple sport-inspired domains. Each group will be assigned one axis and will collaboratively develop benchmark specifications and concrete evaluation protocols. Candidate breakout axes include:

- **Coordination across timescales:** Benchmarks for collaboration across fast reactions, between-play adaptation, and longer-horizon strategy.
- **Physical coupling and contact:** Benchmarks for shared-object interaction, physical proximity, contact, and recovery from unexpected interaction.
- **Communication and intent inference:** Benchmarks for verbal, gestural, gaze-based, and implicit communication during fast physical collaboration.
- **Role switching and initiative-taking:** Benchmarks for dynamic transitions between leader, follower, supporter, defender, and initiator roles.
- **Safety, trust, and graceful failure:** Benchmarks for physical safety, perceived safety, trust calibration, and safe degradation under failure.
- **Morphology and fairness:** Benchmarks for comparing heterogeneous robot embodiments and defining fair evaluation rules across platforms.

Groups will collaborate on the assigned axis for 20 minutes and then present their outcomes in brief 2-minute pitches to the full workshop audience. A breakout discussion facilitator handout, included in the Appendix, will define the assigned axes in more detail, structure the discussion,

record benchmark specifications, and guide the final group report-out. This structured collaborative format will encourage goal-oriented interaction between participants and produce concrete benchmark design principles for agile physical human-robot interaction.

Invited Talks, Moderated Audience Discussion: Talks will feature experts from academia, industry, and early-career researchers, representing the state-of-the-art in athletic robots, embodiment design, and HRI, with dedicated Q&A sessions to encourage direct audience engagement. A moderated discussion session will follow, where a moderator will invite audience to ask questions, share perspectives, and explore contrasting viewpoints. To facilitate participation, we will set up an online portal for anonymous submission of questions and comments throughout the session, which organizers will relay to speakers and moderators.

Panel Discussion: Invited speakers will discuss key questions towards forming a community roadmap for agile physical HRI, such as benchmarks for advancing mixed human-robot sports, and translation of collaborative athletic robotics into real-world applications. We will include time for questions from the audience to provide an open platform for discussion.

Spotlight Talks, Poster Session, Robot Demonstration Videos: Students and early career researchers will present their work in short spotlight presentations, followed by a poster session. Participants will have the opportunity to share their research and receive feedback through direct interaction with attendees, fostering discussion around emerging ideas and open research challenges. Additionally, videos of robots demonstrating agility and collaboration will be curated from the internet and played during the poster session. We expect the visual presentation to improve and ground the focus of discussion among poster session attendees.

Benchmark Prototype: As a concrete workshop outcome, we will develop a community-informed simulation benchmark prototype for mixed human-robot sports, based on doubles tennis. The prototype, seeded by our prior wheelchair tennis research, will be refined through input collected during the breakout sessions, moderated discussions, and panel discussion. The goal is not to prescribe a single benchmark in advance, but to use the workshop to identify reusable design principles for sport-inspired simulation benchmarks, including task abstractions, difficulty progressions, safety constraints, fairness criteria, and evaluation metrics.

The doubles tennis benchmark will consider the following progressively rich levels. Videos of baseline behaviors are available on the website. Workshop feedback will be used define both objective metrics, such as task success and game score, and human-centered metrics, such as perceived safety, workload, and enjoyment.

1. **Kinematic Interception:** The agent must navigate to place an incoming ball within the racket workspace.
2. **Dynamic Returns:** The agent must return the ball to the opponent’s side considering actuator, ball-flight, and racket-ball contact dynamics.
3. **Collaborative Rally:** Human/robot collaborators are introduced in 1+1 or 2+2 configurations and success is measured by safe sustained rallying. This level supports two settings: (1) policies for all agents are learned jointly (2) policies for collaborators are predefined.
4. **Competitive:** The agent must compete and defeat the opponent in a 1v1 setting with a standard tennis scoring scheme. Opponent policies of various proficiencies are provided.
5. **Mixed Collaboration-Competition:** The agent must collaborate with its teammate and defeat the opponent team in a 2v2 configuration. Opponent team policies of various proficien-

cies are provided. This level supports two settings: (1) policies for the agent and teammate are learned jointly (2) policies for the collaborating teammate are predefined.

Predefined behaviors for proxy-human agents will be initially realized with scripted, parameterized policies, baseline learning/planning agents, and in the long-term support richer community-developed human behavior models.

Contributed Papers Plan

We will solicit contributed papers through targeted invitations and broad community outreach, including authors of recent relevant work, robotics mailing lists, industry and university groups, social media, and the network developed through previous RoboLetics workshops. We will invite non-archival submissions of up to four pages (excluding references) in the CoRL format, emphasizing original in-progress research, position papers, benchmark proposals, and system demonstrations. Submissions will be managed through OpenReview with submission, review and notification timelines advertised on the workshop website.

Each submission will receive at least two independent reviews from qualified researchers, with decisions based on relevance, originality, technical quality, and potential to stimulate discussion. We will not set a fixed acceptance rate. Large language models will not be allowed for generating reviews, to preserve review integrity. Accepted papers will be presented as spotlight talks and posters. A best paper award will recognize an outstanding contribution through a cash award sponsored by the Robotics and AI Institute.

Plans to Encourage Participation and Diversity

The workshop is designed to encourage active participation and meaningful discussion among attendees across all career stages. Student spotlight presentations and interactive poster sessions will provide opportunities for early-career researchers to receive feedback from experts in the field. Talks will provide time for questions and engagement with invited speakers. We additionally designate moderated discussion periods following invited talks to allow open-ended discourse among audiences on key themes. Breakout sessions are designed to encourage collaboration between diverse participants towards preparing short proposals that identify key research gaps. Finally, the workshop will conclude with a panel discussion bringing together invited speakers from academia and industry towards outlining a community roadmap for mixed human-robot sports based benchmarks.

Logistics and Resources

Estimated audience size: 40-50 (in-person)

Physical requirements: We request a projector and a screen to display workshop materials, a podium for speakers, 3-4 microphones for moderators, speakers and audience, power supply for organizers' computers, easels and poster-boards for 10-15 posters, 7-8 whiteboards with markers for breakout sessions, and seating arrangements for in-person attendees.

Special requirements: None.

Commitments

- ✓ At least one organizer will attend CoRL 2026 in person.
- ✓ Organizers will attend the CoRL 2026 workshop organizers meeting.
- ✓ Organizers will submit a post-workshop artifact.
- ✓ Willingness to merge with related proposals if requested.

Our preference is to merge if the focus of the workshop is preserved.

Appendix: Breakout Discussion Facilitator Handout

Goal. Each group will develop benchmark specifications for one core axis of agile physical human-robot interaction. The objective is to identify benchmark design principles that use sport-inspired interaction properties to stress-test capabilities relevant to agile, safe, and collaborative human-robot systems.

Axis guide. Each group will be assigned one of the following axes. The descriptions below are intended to help groups interpret the assigned axis and translate it into benchmark specifications.

- **Coordination across timescales:** Focuses on collaboration across fast reactions, between-play adaptation, and longer-horizon strategy.
- **Physical coupling and contact:** Focuses on shared-object interaction, physical proximity, contact, and recovery from unexpected interaction.
- **Communication and intent inference:** Focuses on verbal, gestural, gaze-based, and implicit communication during fast physical collaboration.
- **Role switching and initiative-taking:** Focuses on dynamic transitions between leader, follower, supporter, defender, and initiator roles.
- **Safety, trust, and graceful failure:** Focuses on physical safety, perceived safety, trust calibration, and safe degradation under failure.
- **Morphology and fairness:** Focuses on comparing heterogeneous robot embodiments and defining fair evaluation rules across platforms.

Assigned axis: _____

Example axes include coordination across timescales; physical coupling and contact; communication and intent inference; role switching and initiative-taking; safety, trust, and graceful failure; and morphology/fairness.

Suggested discussion structure for 20 minutes

- **Minutes 0–3:** Clarify the assigned axis and choose 1–2 sport-inspired examples to ground the discussion.
- **Minutes 3–10:** Define the target capability and minimal benchmark task abstraction.
- **Minutes 10–15:** Specify progressive difficulty levels, safety/fairness constraints, and evaluation metrics.
- **Minutes 15–20:** Identify failure modes, open questions, and prepare a 2-minute report-out.

Benchmark design worksheet

1. Target capability being evaluated

For the assigned axis, what specific capability should the benchmark evaluate? Define the capability in a way that could apply across multiple sport-inspired domains and robot embodiments.

2. Sport-inspired interaction properties

What sport-inspired interaction properties make this axis challenging? Consider properties such as speed, uncertainty, physical proximity, cooperation, competition, shared objects, contact, or strategic pressure.

3. Minimal benchmark task abstraction

What is the simplest benchmark task that captures the assigned axis? Define the agents, environment, objective, available observations/actions, and success criteria.

4. Progressive difficulty levels

How should the benchmark become harder while still isolating the assigned axis? Consider increasing speed, uncertainty, physical proximity, interaction complexity, opponent behavior, communication burden, or strategic depth.

5. Safety and fairness constraints

What constraints are needed to evaluate the assigned axis safely and fairly? Consider physical safety, perceived safety, stopping conditions, force/speed limits, protected zones, human comfort, and morphology-specific advantages.

6. Quantitative and qualitative metrics

What measurements would show whether a robot performs well on this axis? Consider task success, timing, robustness, coordination quality, human effort, perceived safety, trust, legibility, and recovery from failure.

7. Expected failure modes and open questions

Where might current systems fail on this axis? What research questions would this benchmark expose for robot learning, control, perception, planning, human modeling, embodiment design, or HRI?

Suggested Format for 2-minute report-out

- **Axis:** What interaction axis did your group focus on?
- **Benchmark idea:** What is the minimal benchmark task abstraction?
- **Progression:** How does the benchmark move from easy to hard?
- **Metrics:** What would you measure?
- **Research gap:** What open problem does this benchmark reveal?

Deliverable: One benchmark design principle that can inform a broader roadmap for agile physical human-robot interaction.